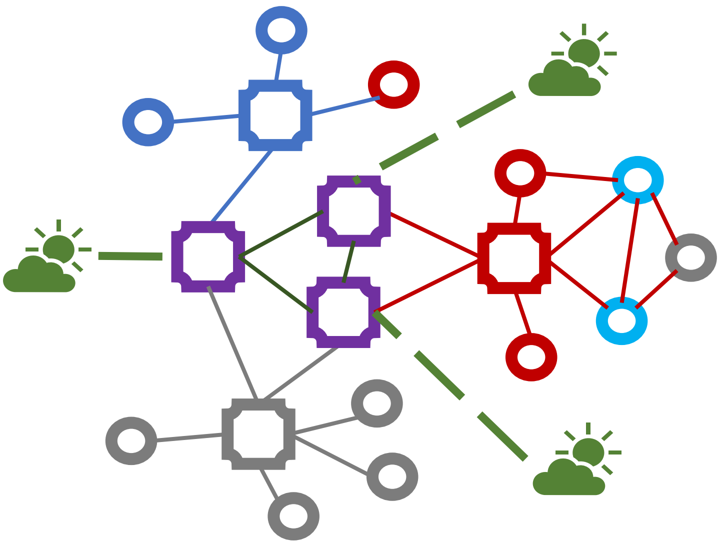
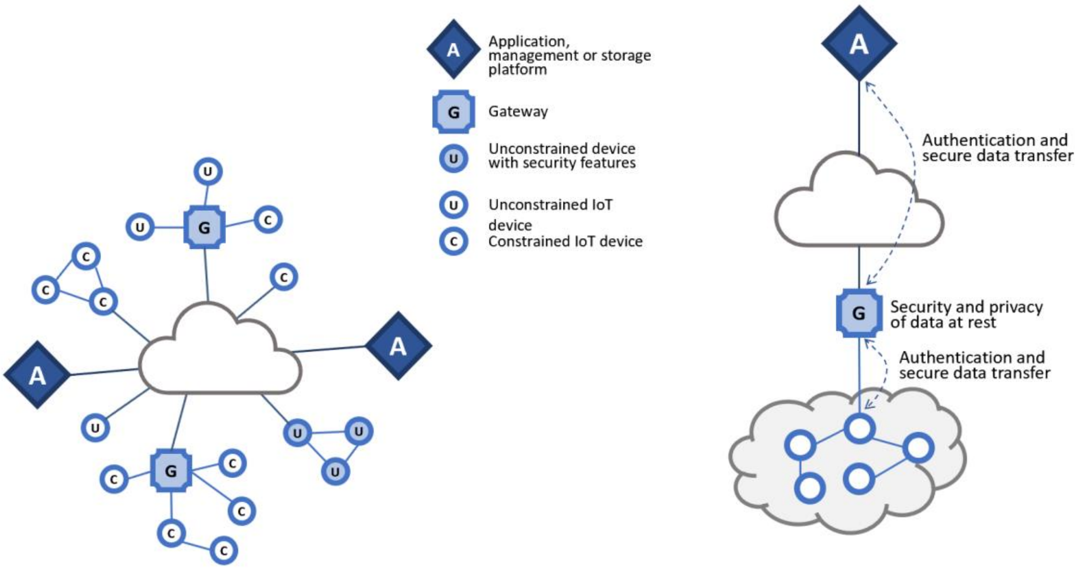
At the oral exam I will start the exam asking you to comment one of these schemas or to answer to one of these questions:

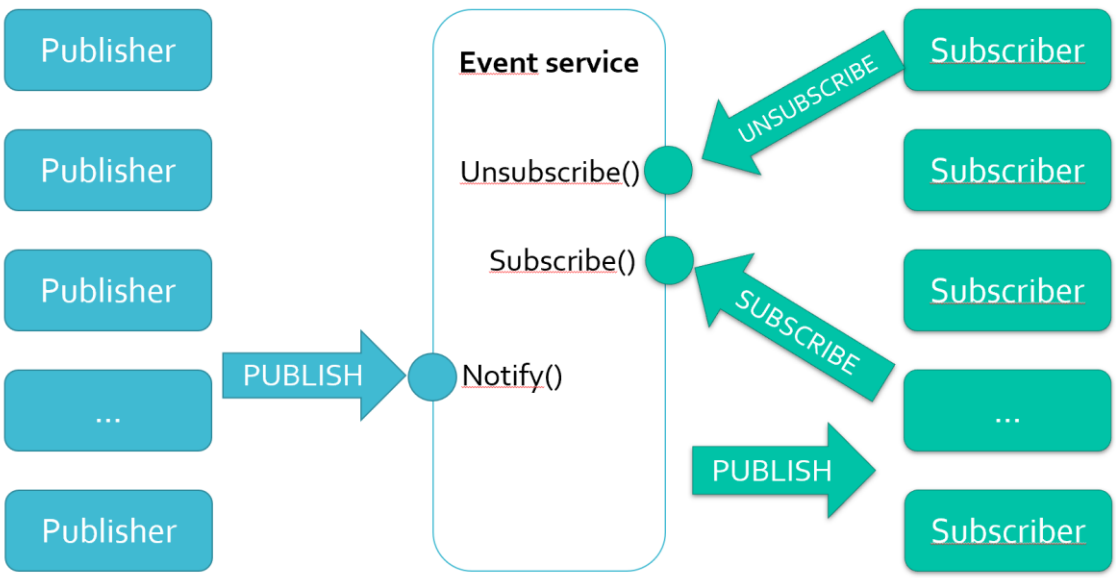
interoperability



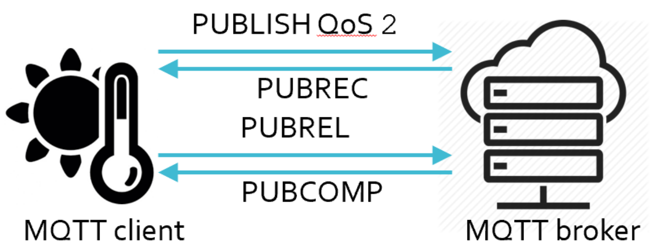
IoT & security



Publish/subscribe & MQTT



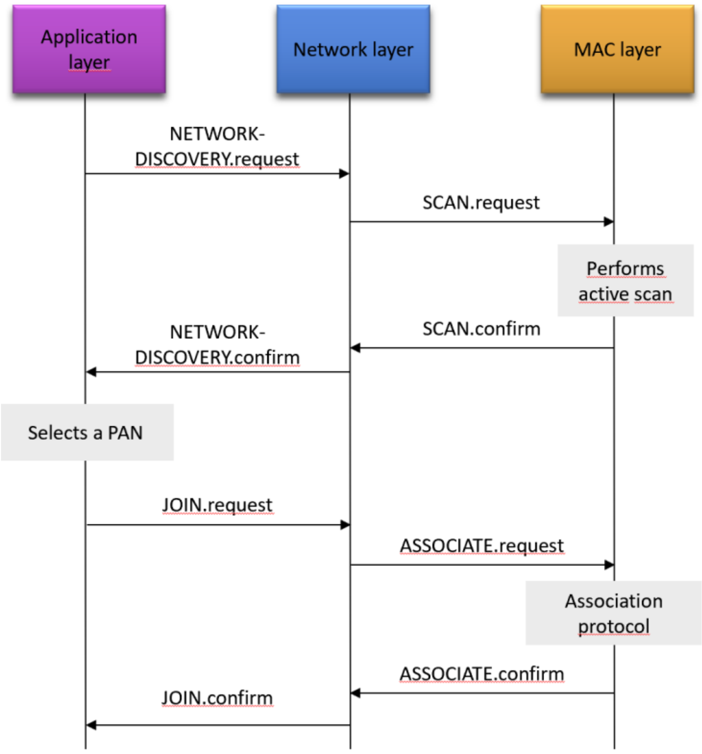
MQTT – II



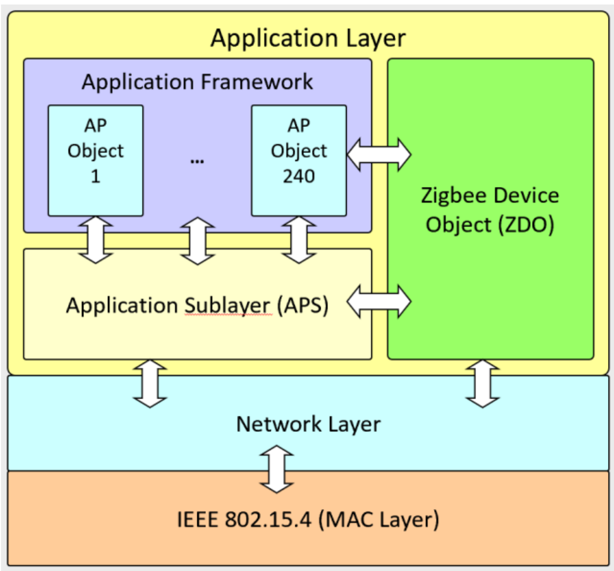
MQTT-III

- Explain how messages are filtered by the broker and what are the topics in MQTT
- Explain the persistent sessions in MQTT
- Explain the retained messages in MQTT
- Explain the Lastwill&testament and the Keep Alive mechanism in MQTT.

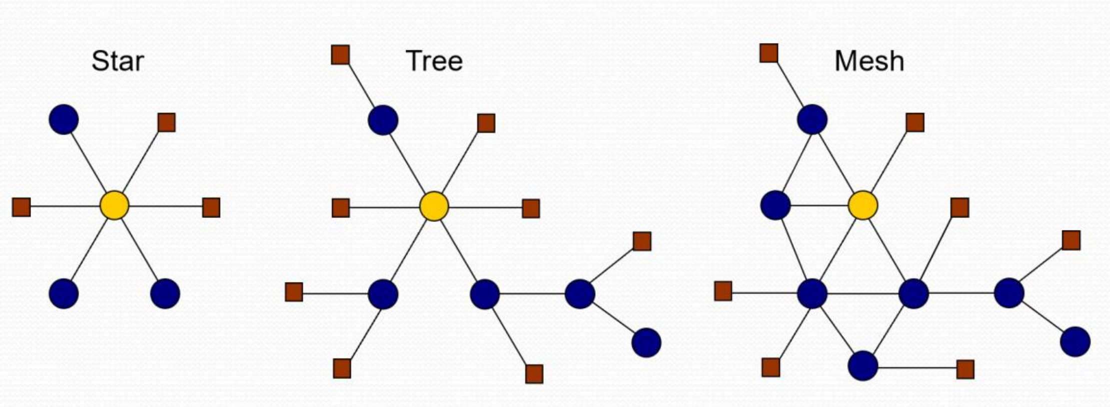
ZigBee - I



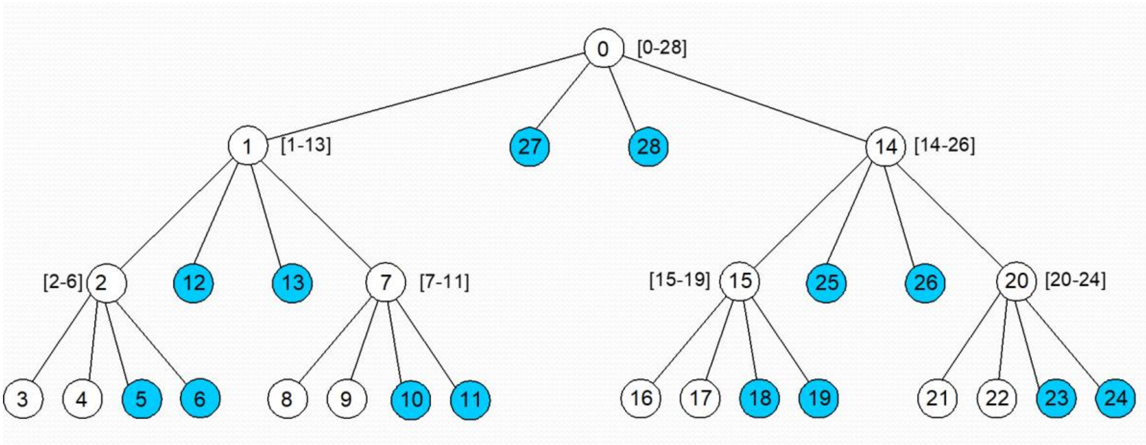
ZigBee – II



Zigbee - III



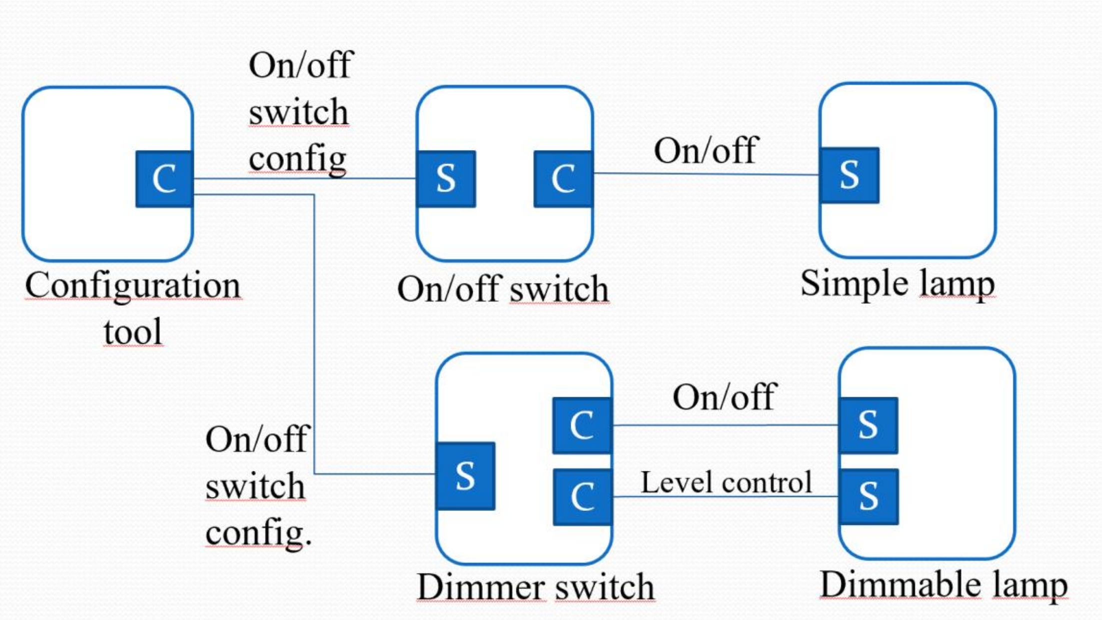
ZigBee - IV



ZigBee - V

Src Addr (64 bits)	Src EP	Cluster ID	Dest Addr (16/64 bits)	Addr/Grp	Dest EP
0x3232...	5	0x0006	0x1234...	A	12
0x3232...	6	0x0006	0x796F...	A	240
0x3232...	5	0x0006	0x9999	G	_
0x3232...	5	0x0006	0x5678...	A	44

ZigBee - VI

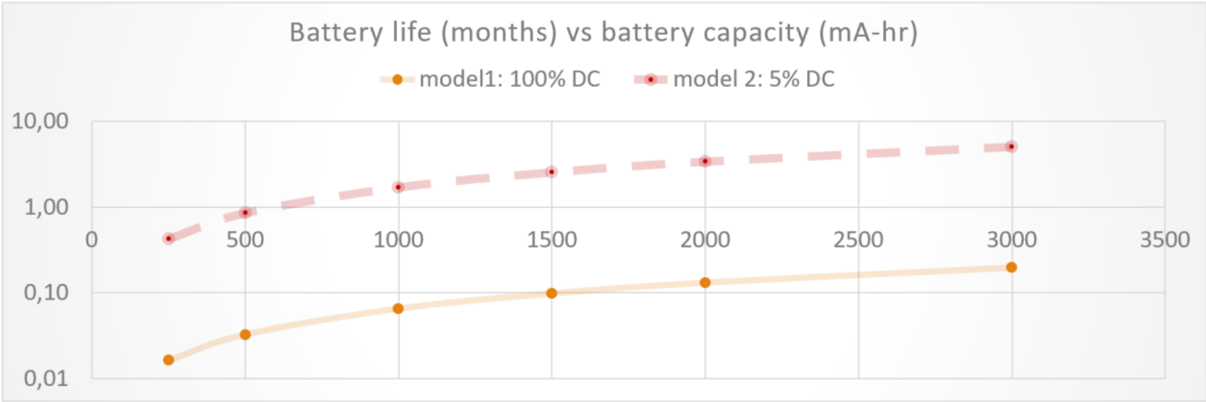


duty cycle - I

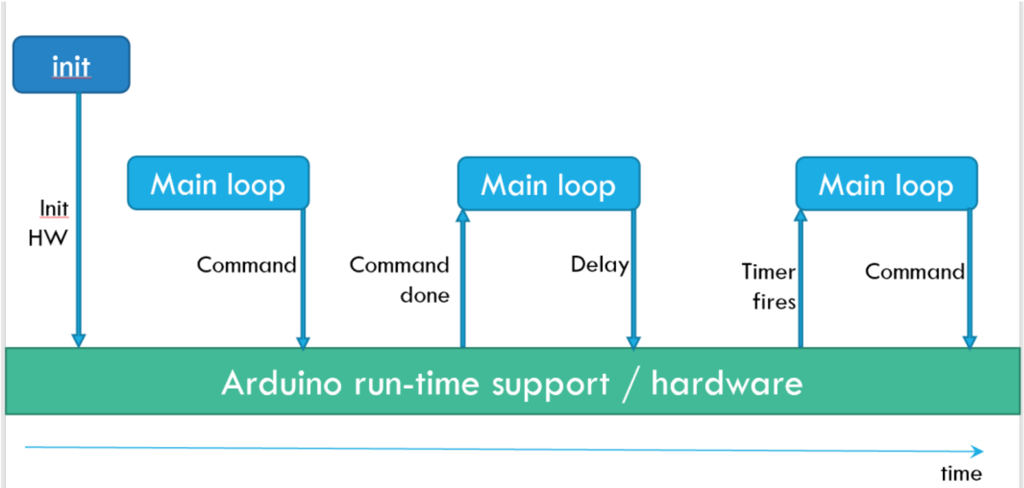
```
void loop() {
 // reads the input from analog pin 0:
 turnOn(analogSensor);
 int sensorValue = analogRead(A0);
 turnOff(analogSensor);
  // converts value into a voltage (0-5V):
  float voltage = sensorValue * (5.0 / 1023.0);

 // transmits voltage over the radio
 turnOn(radioInterface);
 Serial.println(voltage);
 turnOff(radioInterface);
 // waits for next loop
 idle(380);
}
```

	Value	units
Micro Processor (Atmega128L)		
current (full operation)	8	mA
current sleep	15	µA
Radio		
current in receive	19,7	mA
current xmit	17,4	mA
current sleep	20	µA
Logger (storage in the flash memory)		
write	15	mA
read	4	mA
sleep	2	µA
Sensor Board		
current (full operation)	5	mA
current sleep	5	µA
Battery Specifcations		
Capacity Loss/Yr	3	%

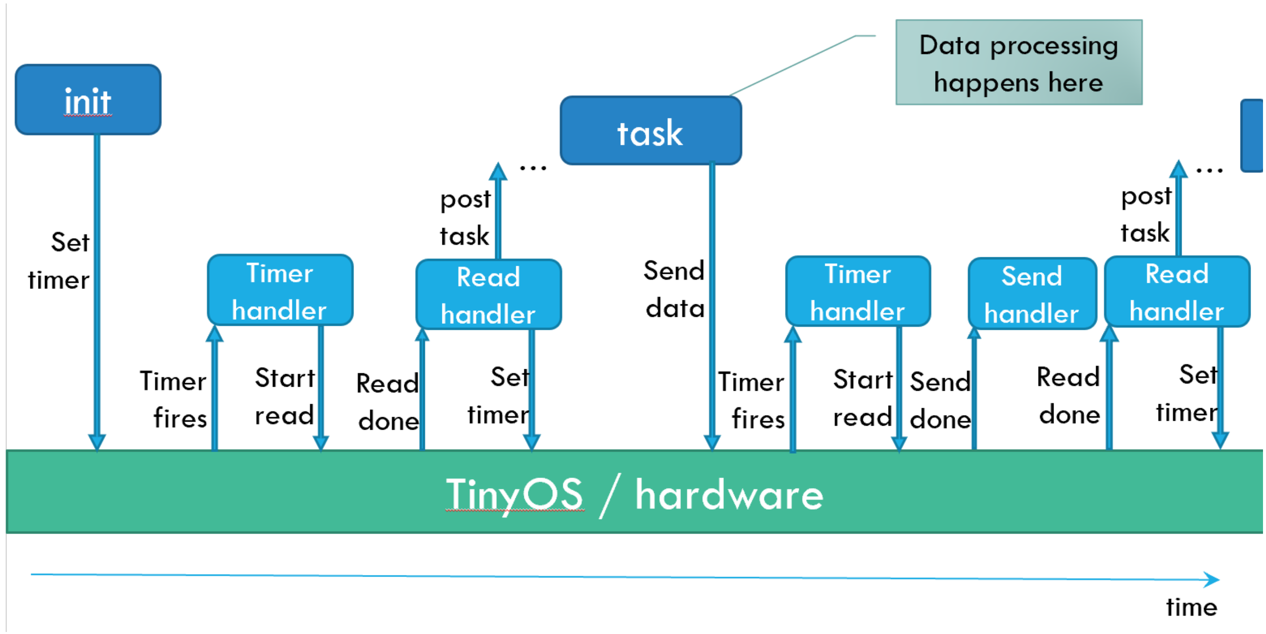
duty cycle - II



embedded programming - I



embedded programming - II

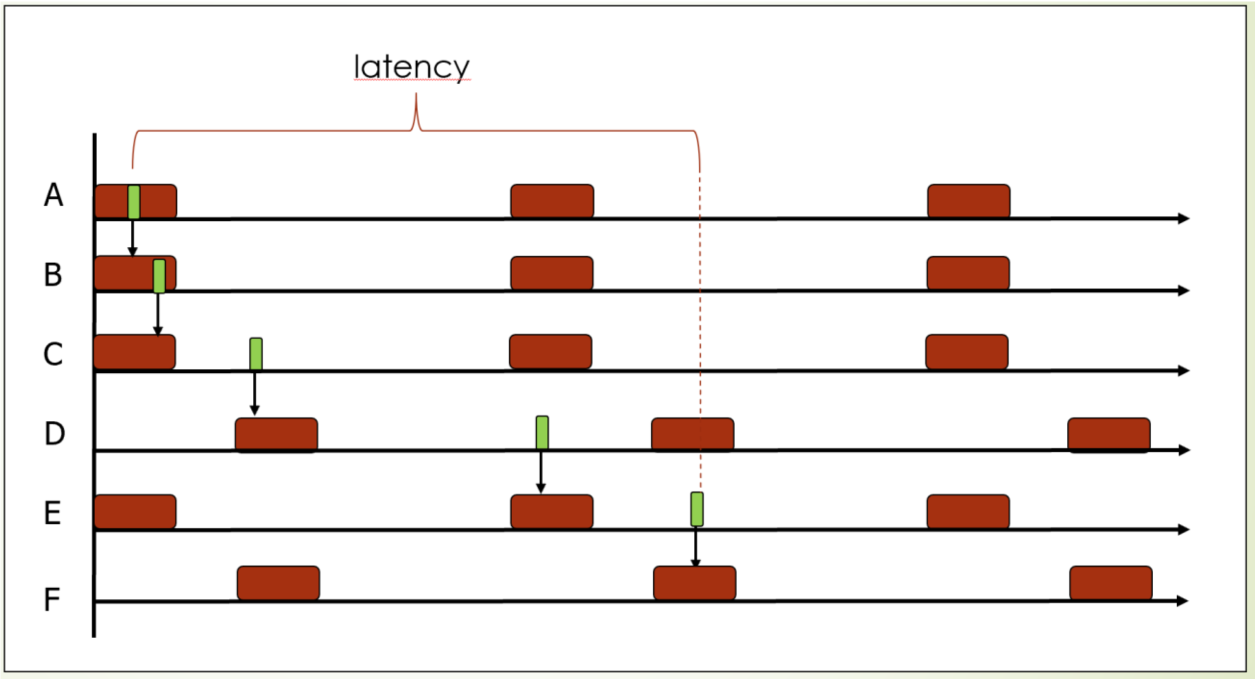


Arduino - interrupt

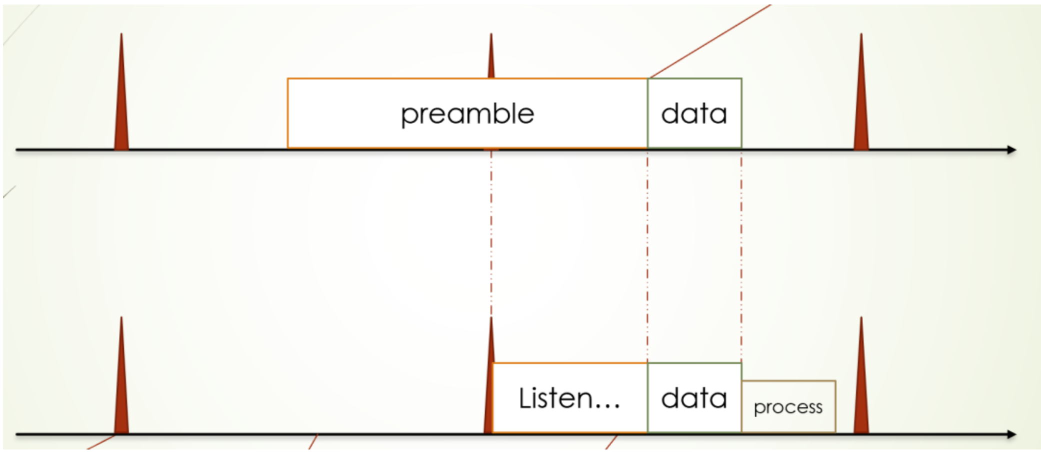
```
/**
 * test interrupt
 *
 */

volatile int greenLed=7;
volatile int count=0;

void setup()
{
  Serial.begin(9600);
  pinMode(greenLed, OUTPUT);

  digitalWrite(greenLed, LOW);

  attachInterrupt(0,
      interruptSwitchGreen,RISING);
  // 0 because it is pin 2
}
```

```
void loop() {
  count++;
  delay(1000);
  // interrupts are received
  // also within delay!

  Serial.print(“waiting:");
  Serial.println(count);

  if ( count == 10 )
  {
    count = 0;
    digitalWrite(greenLed, LOW);
    Serial.println(“now off");
  }
}

void interruptSwitchGreen()
{
  digitalWrite(greenLed, HIGH);
  count=0;
  Serial.print(“now on");
}
```

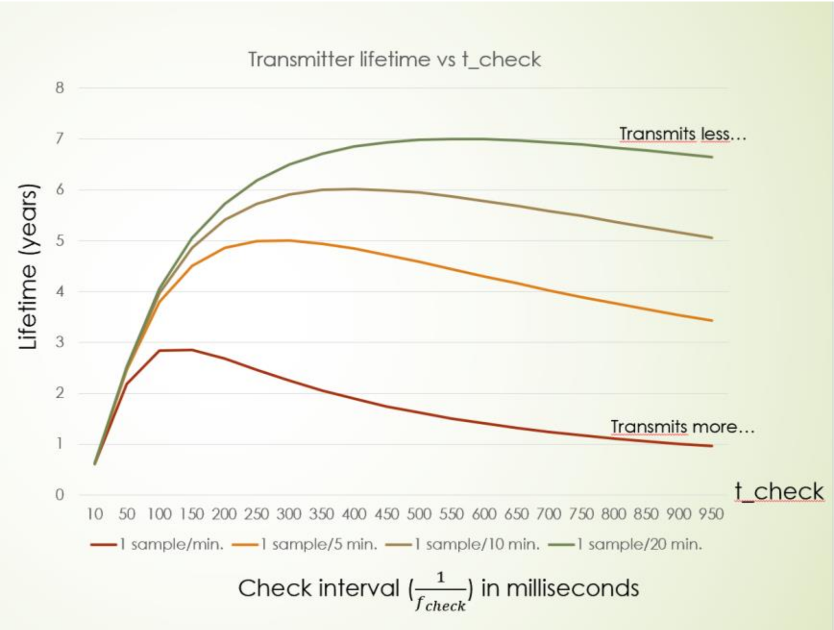
SMAC



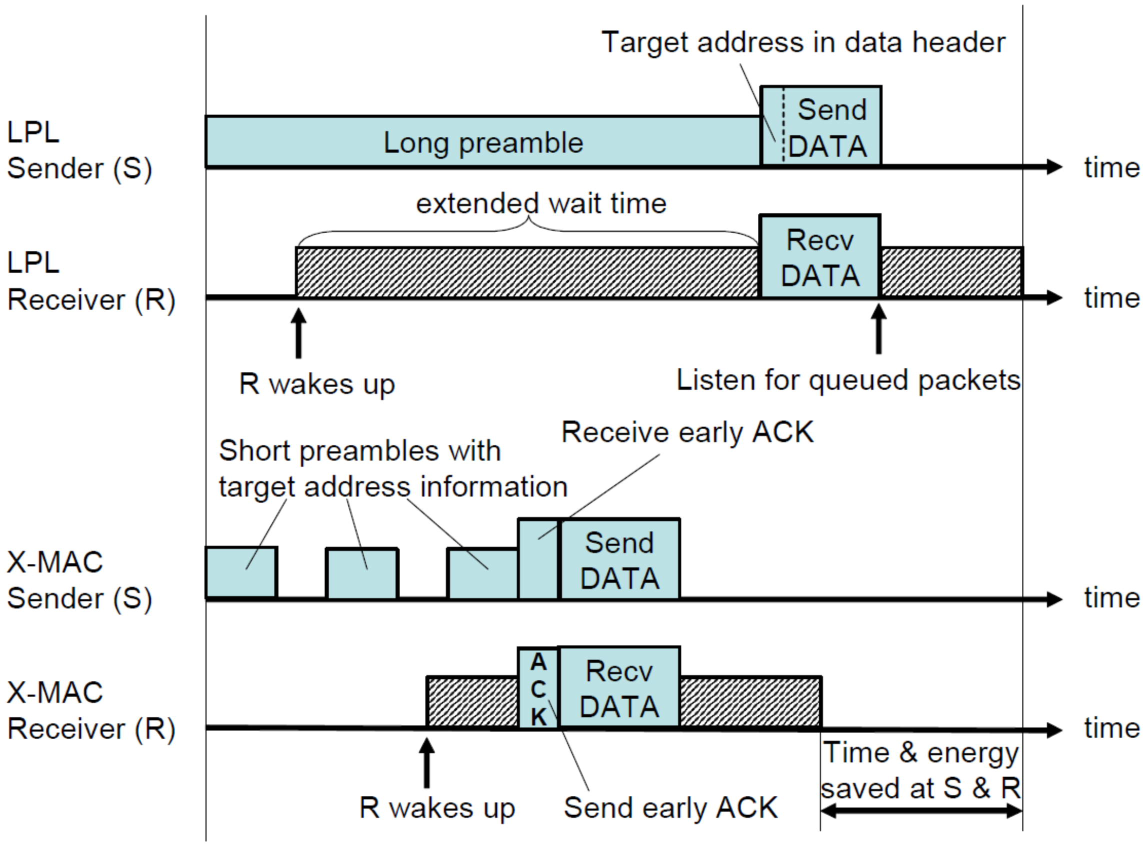
BMAC-I



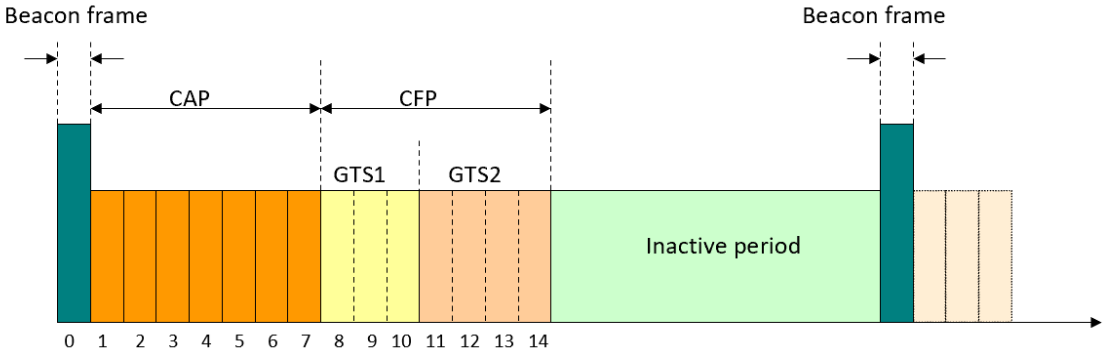
BMAC-II



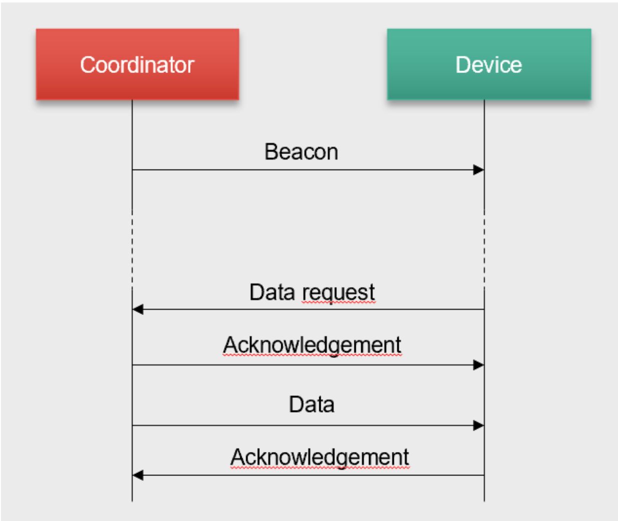
X-MAC/B-MAC



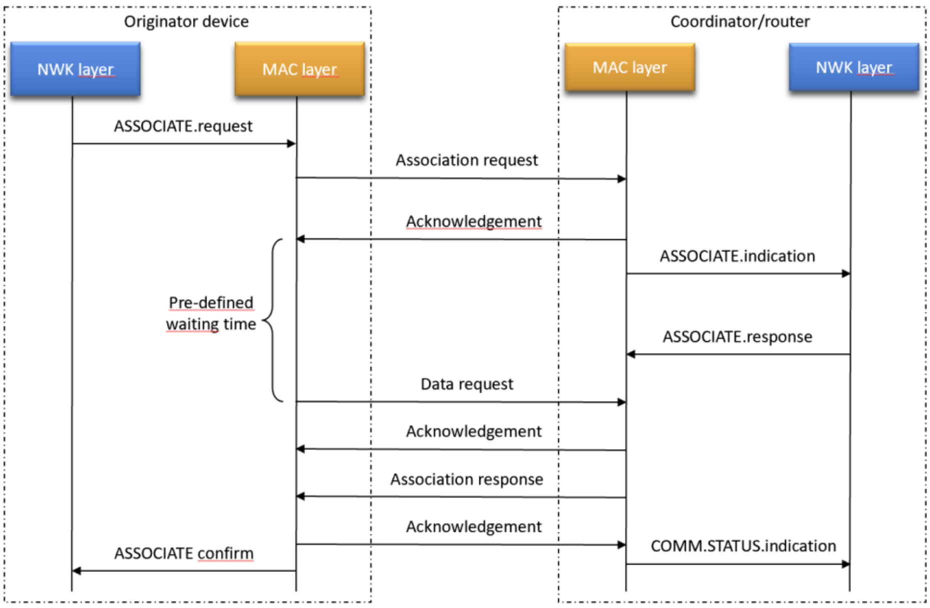
IEEE 802.15.4 - I



IEEE 802.15.4 - II



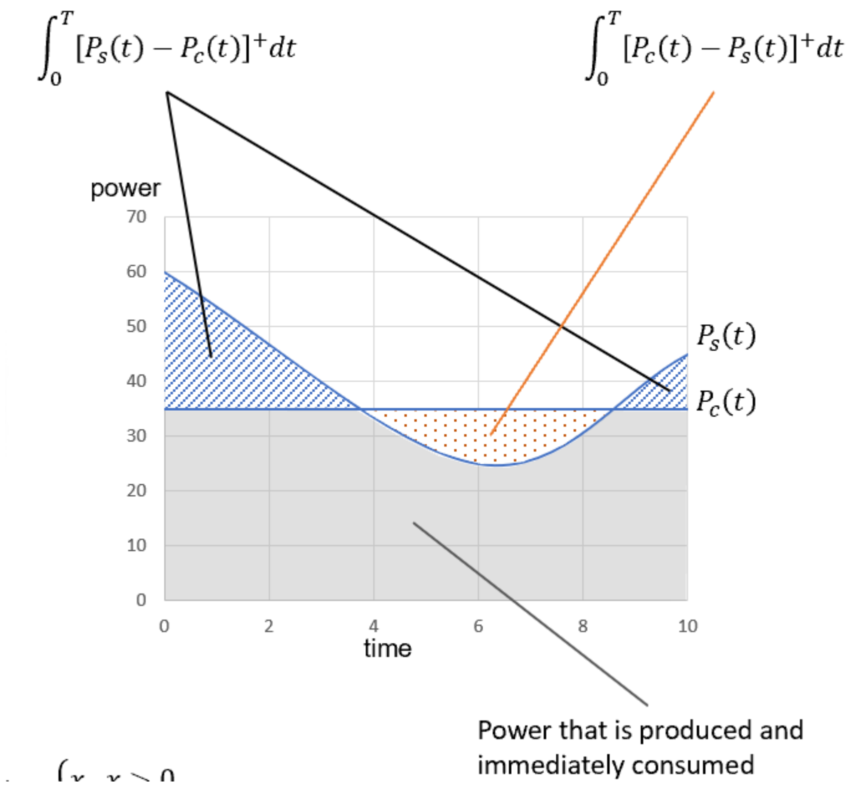
IEEE 802.15.4 - III



Energy harvesting

- Explain the difference between an harvest-use and an harvest-store-use architecture
- Explain the classification of sources in terms of controllability and predictability
- Explain the concept of energy neutrality

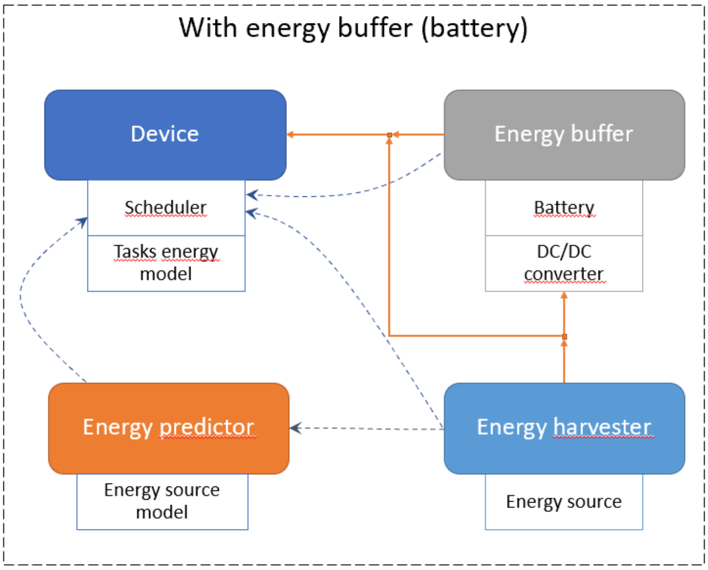
Energy harvesting - I



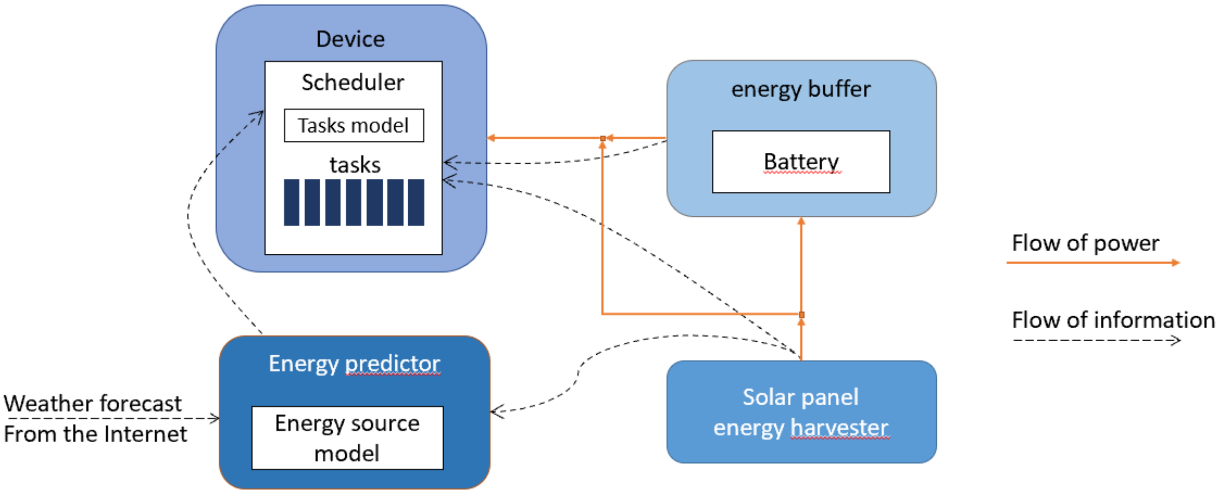
Energy harvesting – II

platform	v_{min} (volts)	v_{max} (volts)	v_{ref} (volts)	quantization levels (bit)	x_{min}	x_{max}	size of battery (mAh)	x (voltage sampling)	Battery charge (mAh)
RPI	3,5	5	0,004883	10	716	1023	3000	1000	2798
Arduino	7	9	0,008789	10	795	1023	3000	800	359
Tmote	1,8	3	0,000732	12	2457	4095	3000	3277	1652

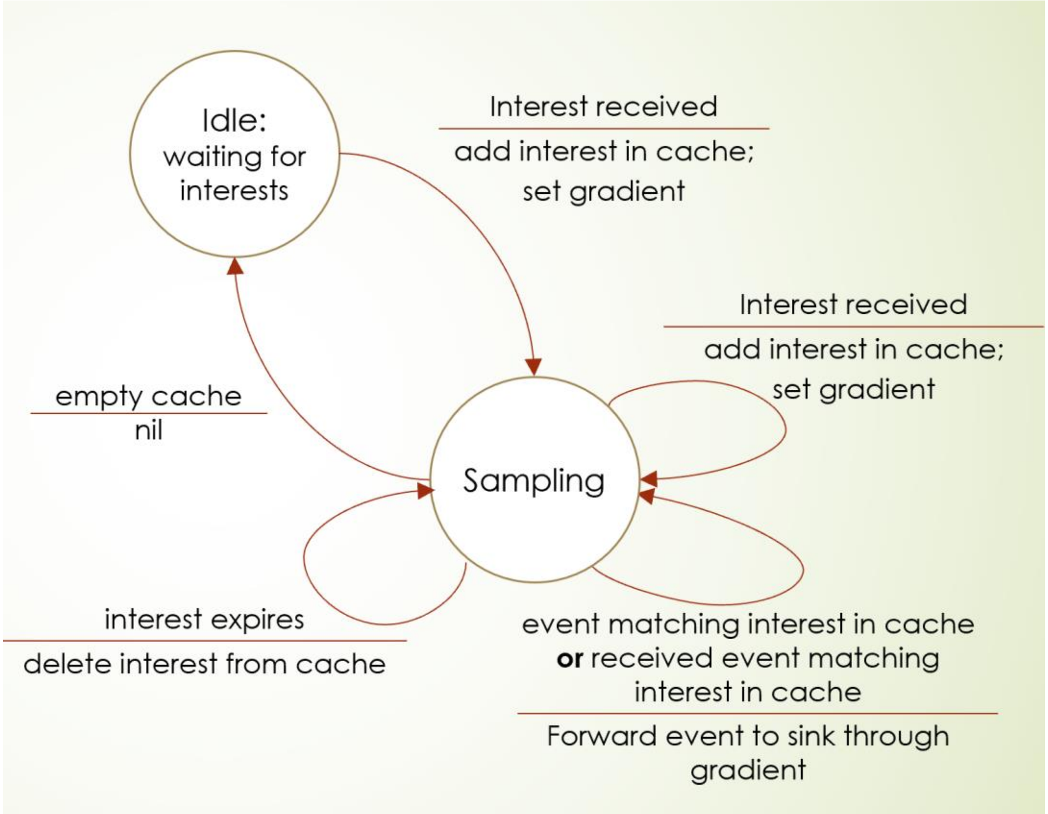
Energy harvesting – III : explain the Kansal approach to energy neutrality



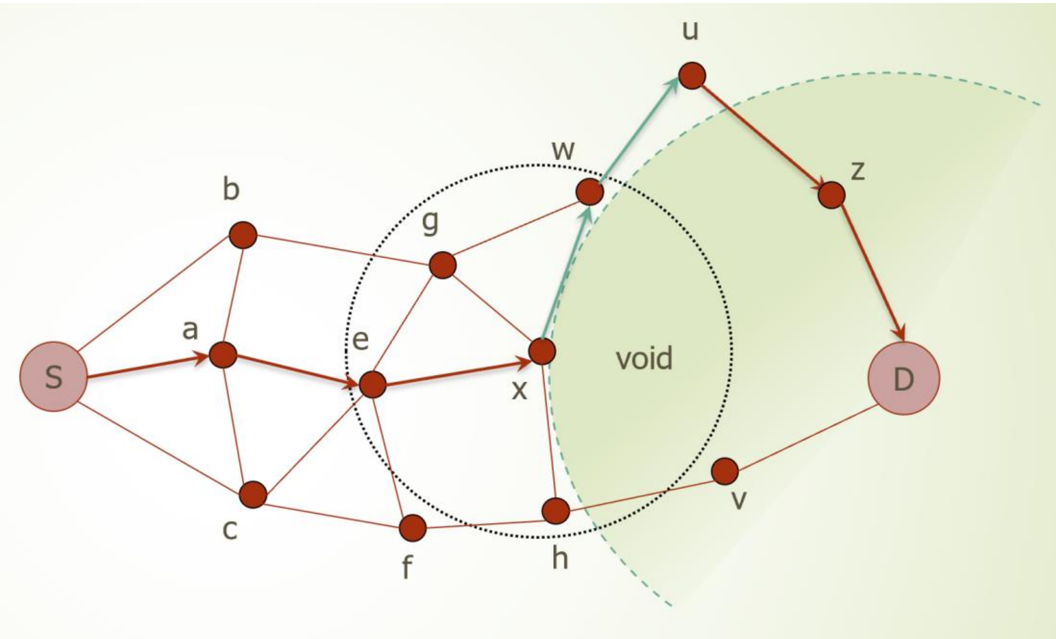
Energy harvesting – IV : explain the task model for energy neutrality



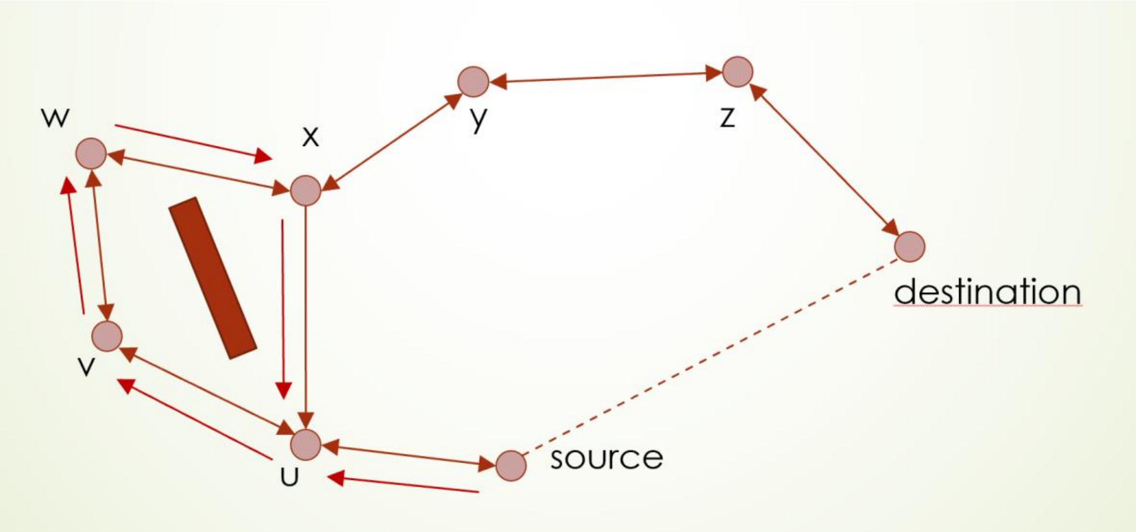
Directed Diffusion



GPSR - I



GPSR - II



GPSR – III